

Section 3.7 Derivatives as Rates of Change

Ex ① The height of A ball thrown into the air is $h(t) = -16t^2 + 90t + 4$ feet After t seconds.

A) Find velocity when $t=1$.

$$h'(t) = -32t + 90$$

$$h'(1) = -32(1) + 90 = 58 \frac{\text{ft}}{\text{sec}}$$

$$h(t) = \text{dist.}$$

$$h'(t) = \text{velocity}$$

$$h''(t) = \text{acceleration}$$

B) Find acceleration when $t=1$.

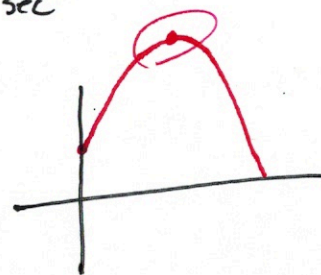
$$h''(t) = -32 \frac{\frac{\text{ft}}{\text{sec}}}{\text{sec}} \text{ or } \frac{\text{ft}}{\text{sec}^2}$$

C) when is velocity zero?

$$0 = -32t + 90$$

$$32t = 90$$

$$t = \frac{90}{32} = 2.813 \text{ sec's}$$



Ex ② The volume of a spherical balloon
is $V = \frac{4}{3} \pi r^3$

Find $\frac{dV}{dr}$ when $r = 1$ foot ; Explain.

$$\frac{dV}{dr} = \frac{4}{\cancel{3}} \pi \cdot \cancel{3} r^2$$

$$\frac{dV}{dr} = 4\pi (1)^2 = 12.566$$

Volume is increasing by 12.566 ft^3
for each 1 foot increase in radius.

Ex (3) The horizontal distance an object has travelled (in feet) is given by

$$s(x) = 3x^3 - 5x^2 - 6x + 4 \text{ with } x \text{ in seconds.}$$

A) Find velocity when $t = 2$.

$$s'(x) = 9x^2 - 10x - 6$$

$$s'(2) = 9(2)^2 - 10(2) - 6$$

$$s'(2) = 36 - 20 - 6 = 10 \frac{\text{ft}}{\text{sec.}}$$

B) When is it at rest?
Velocity = 0

$$0 = 9x^2 - 10x - 6$$

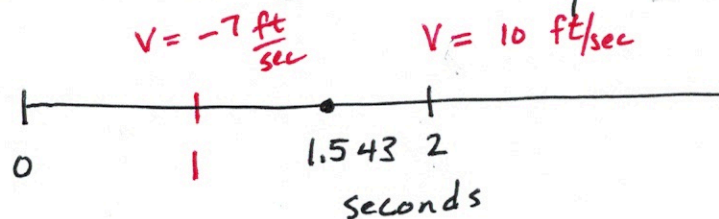
$$x = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(9)(-6)}}{2(9)}$$

$$x = \frac{10 \pm \sqrt{316}}{18}$$

$$x = 1.543 \text{ sec's}$$

Ex(3)

c) when is it going in the positive direction?



$$\begin{aligned} s'(1) &= 9(1)^2 - 10(1) - 6 \\ &= 9 - 16 = -7 \frac{\text{ft}}{\text{sec}} \end{aligned}$$

(1.543, ∞) positive direction

D) Find Acceleration when $t = 2$.

$$s'(x) = 9x^2 - 10x - 6$$

$$s''(x) = 18x - 10$$

$$s''(2) = 18(2) - 10$$

$$= 36 - 10 =$$

$$\boxed{26 \frac{\text{ft}}{\text{sec}^2}}$$